

RICHMOND ESSIEKU

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EDUCATION

Texas Tech University, Lubbock, TX

Ph.D. in Economics

Expected June 2026

University of Texas Rio Grande Valley, Edinburg, TX

MSc. in Applied Statistics and Data Science

August 2022

Xiamen University, Xiamen, China

MSc. in Applied Economics

May 2019

University of Cape Coast, Cape Coast, Ghana

BSc. in Mathematics with Economics

May 2016

FIELDS OF INTEREST

Health, Labor, and Household Economics; Policy Evaluation; Applied Econometrics (causal inference & AI/ML)

RESEARCH

Published Article

- “Stability analyses on the effect of vaccination and contact tracing in monkeypox virus transmission.” Published in *J. Math. Comput. Sci.*, 13 (2023), Article ID 8, (Joint with Solomon Eshun and James Ladzekpo). [View Paper](#)

Working Papers

- **Diagnosed Too Soon or Too Late? Labor Market and Health Outcomes of Adults with Early vs. Late ADHD Diagnosis** ([Job Market Paper](#)). Published in *SSRN*, 2025. [View Paper](#)

Abstract: The study examines whether the timing of ADHD diagnosis matters. Using nationally representative U.S. Medical Expenditure Panel Survey data from 2008–2022 and a within-diagnosed design, I estimate average treatment effects with the augmented inverse probability weighting and corroborate the findings with propensity score matching, instrumental-variable estimates, and double machine learning. Earlier diagnosis delivers durable long-term health gains, raises work hours by 13.3%, and increases years of schooling, yet average labor income lags relative to later-diagnosed counterparts, with the wage penalty larger among those with more severe symptoms. A mechanism-consistent account emerges from wage equations and a Blinder–Oaxaca decomposition: Mincer marginal returns are lower for the early-diagnosed (schooling about 4.6% versus 9.4%; experience about 3.3% versus 9.2%), and the pricing of the same endowments is adverse for those diagnosed early. Timing matters: effects are strongest at ages 5–6 (work hours rise to 15.7%; health improvements largest) and remain stable through age 12. The findings motivate pairing very early screening and integrated healthcare interventions with labor-market tools that raise realized wage returns and mitigate structural barriers. Policy implications are discussed.

- **Labor Market Margins and the Affordable Care Act’s Health Insurance Reforms: Racial Heterogeneity Among Pre-Retirees**. Published in *SSRN*, 2025. [View Paper](#)

Abstract: This study examines the causal effects of the ACA’s health insurance reforms on labor market extensive and intensive margins among pre-retirees, focusing on racial disparity. Using a double robust method of propensity score matching with ATT and a staggered difference-in-differences design, we estimate the impact of Health Insurance Marketplace Subsidies (HIMS) and Medicaid expansion. Our findings indicate that HIMS delays retirement and prevents labor force exit (Number Needed to Treat, NNT, equals 132), meaning that for every 132 individuals subsidized, one additional person stays employed who would have otherwise exited. On the intensive margin, HIMS increases part-time work by 1.51 percentage points (NNT equals 66). Weekly hours drop by 1.24.

Racial disparities emerge, with White and other racial groups (Asian) benefiting most, while Black and Hispanic workers experience weaker effects. No significant labor market effects are observed from Medicaid expansion.

- **Mortality Risk Under Occupational and Health Comorbidity: Double Machine Learning Assessment of the Obesity–Diabetes Double Burden.** Published in *SSRN*, 2025. [View Paper](#)

Abstract: This paper estimates the impact of type 2 diabetes on mortality among obese adults in the United States, addressing a key gap in the economics of chronic illness. Leveraging nationally representative data from the National Health Interview Survey (NHIS) linked to mortality records, we apply double machine learning to estimate the marginal impact of diabetes conditional on obesity. Results indicate that among obese individuals, diabetes increases the probability of all-cause mortality. Subgroup analyses show that this risk is especially pronounced among males, adults aged 60 to 74, individuals with hypertension, and those employed in physically demanding or public-sector occupations. A policy-relevant calculation suggests that for every 29 obese adults who avoid or delay the onset of diabetes, one premature death could be prevented. A random forest algorithm analysis highlights insurance status, educational attainment, and poverty level as dominant predictors of mortality risk, reinforcing the importance of social and economic context in shaping health outcomes.

- **Predicting Obesity Risk Through Gene-Level Modeling: A Machine Learning Approach to Inform Precision Public Health Policy.** [View Abstract on poster # 11](#)

Excerpt: “Can gene-level signals help us get ahead of the obesity epidemic? Leveraging DNA methylation and HPA-axis–focused feature discovery (SAM + RFE-SVM), this study builds a multilayer perceptron machine learning model that predicts individual obesity risk with strong out-of-sample performance (~83% accuracy, 90% recall, 75% AUC). The results show how gene-based risk stratification can sharpen prevention, target resources, and inform cost-effective, precision public health policy.” (Joint with Rodrigo Hansapani and Vincent Diego).

- **“Labor Force Behavior and ACA Marketplace Premium Subsidy: Margins of Employment Among Older Pre-Retirees”**, Under review at *Health Economics*. [View Paper](#)
- **“Who Benefits from Residential Solar? Evidence on Energy Insecurity Disparities and Policy Gaps in the U.S.”**, Under review at *Energy Policy*. [View Paper](#)
- **“In Whom We Trust? Social Networks and Trust in AI vs. Human Experts in Housing Investment Decisions”** (Joint with Prince Bosompim and Jodi Letikiewicz), Revise and Resubmit at *Finance Research Letters*. [View Paper](#)
- **“When Assistance Isn’t Enough: Energy Insecurity Among Older Adults and Policy Considerations for Energy Assistance Programs”**, Rejected and Resubmit at *Applied Economics*. [View Paper](#)
- **“Buy Now, Struggle Later? Generational Differences in BNPL Use and Household Financial Behaviors”** (Joint with Prince Bosompim), Revise and Resubmit at *Journal of International Banking*.

Work in Progress

- “Machine Learning Signals for Long Term Care Insurance Choice Among Older Americans” (Joint with Francis Essieku)
- “AI or Human Expert? Who Do Financial App Users Trust for Market Forecasts?” (Joint with Prince Bosompim and Gary Mottola)
- “Targeting What Works: AI Driven Clusters Inform Medication, Lifestyle, Family Context, and Workplace Supports for ADHD”
- “Multiple Job Holding and Functional Health Outcomes Across Demographic Lines”

Book Chapter

- “Machine Learning Feature Selection Techniques to Model the Elements of Cash Conversion Cycle and Other Co-

variates on Hospital Performance.” Published in *B.P. International*, 2024. [View Book](#)

TEACHING

Texas Tech University

Teaching Assistant: Recitation Session and Lecture Support

- ECO 2301-001: Principles of Microeconomics Fall 2025
- ECO 4322: The Economics of Labor Markets Spring 2025
- ECO 2302: Principles of Economics II: Macroeconomics Summer 2024
- ECO 3311: Intermediate Macroeconomics Fall 2022, Spring 2024
- ECO 2305: Principles of Economics Fall 2023
- ECO 3312: Intermediate Economic Theory Spring 2023

University of Texas Rio Grande Valley

Instructor

- MATH 1314: College Algebra Spring 2021
- MATH 1301: Statistics and Mathematics — Jump Start Summer 2022

Teaching Assistant: Recitation Session and Lecture Support

- STAT 2334: Applied Statistics for Health Fall 2021
- MATH 1460: Calculus I Summer 2021
- MATH 2412: Pre-Calculus Spring 2022

Regional Maritime University

Teaching Assistant: Recitation Session and Lecture Support

- BLM 105, BPS 105, DPS 105: Elements of Economics Semester 2, 2016
- BLM 201, BPS 201, DPS 201: Principles of Economics Semester 1, 2017
- BLM 101, BPS 101, DPS 101: Business Mathematics Semester 1, 2020

WORK EXPERIENCE

Ghana Revenue Authority, Accra-Tema, Ghana

Statistician and Analyst

November 2015 – July 2016

- Produced high-frequency statistical reports on revenue mobilization and public finance operations, applying economic reasoning to inform tax policy and compliance monitoring.
- Conducted data-driven evaluations of taxpayer behavior and revenue trends contributing to internal auditing processes and econometric insights for decision support.

ACADEMIC SERVICES

Academic Journal Reviewer

2023 – 2025

- Completed 15 peer reviews for the following journals: *International Economics and Economic Policy*, *Journal of Data Analysis and Information Processing (JDAIP)*, *Journal of Service Science and Management*, *iBusiness*.

Conference Abstract Reviewer

2024 – 2025

- Reviewed abstract submissions for academic conferences including: Southwest Decision Science Institute (SWDSI) Annual Conference (2025), IntelliSys Conference (2024–2025), IACMR Conference (2025).

Academic Judge – Student Research and Presentation Competitions

2023 – 2025

- Served as judge for various competitions at Texas Tech University, including: 3-Minute Thesis Competition (2023–2024), Art and Humanities Conference (2023–2024), Undergraduate Research Competition (2025), Graduate Student Research Poster Competition – Abstract Committee (2024).

CONFERENCES AND PRESENTATIONS

- Southern Economic Association 95th Annual Meeting 2025 (scheduled)
- The Washington DC Conference on the Social Sciences 2026 (scheduled)
- Administrative Data Research Conference 2025
- International Forum on Research Excellence, Sigma Xi 2025
- World Obesity and Weight Management Congress 2024
- Obesity Research Institute Annual Conference 2023, 2024
- College of Sciences Research Conference, UTRGV 2024
- West Texas Health Disparities Symposium, TTU 2024
- Graduate Student Research Poster Competition, TTU 2023, 2024

SCHOLARSHIPS AND AWARDS

Academic Scholarships and Fellowships

- Helen Devitt Jones Fellowship Grant, Texas Tech University 2022–Present
- Graduate Assistantship, Department of Economics, Texas Tech University 2022–Present
- Graduate Assistantship, Department of Mathematics and Statistics, UTRGV 2021–2022
- The Washington DC Conference on the Social Sciences academic travel grant 2025
- Department of Economics - Texas Tech University academic travel grant 2025
- Chinese Government Council Scholarship 2017–2019
- HEERF Grant, State of Texas 2021

Research and Competition Awards

- First Place, 22nd Texas Tech Graduate Student Research Poster Presentation 2023
- First Place, Kaggle Data Mining Competition (Team won over 11 competing teams) 2022

PROFESSIONAL MEMBERSHIP

- American Economic Association (AEA)
- American Society of Health Economists (ASHEcon)
- Southern Economic Association (SEA)
- Sigma Xi – The Scientific Research Honor Society

COMPUTER SKILLS

• R • Python • SAS • STATA • LaTeX

REFERENCES

Professor Kaj Gittings

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Texas Tech University
Department of Economics
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Dr. SieWon Kim

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Texas Tech University
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